

Darshan Rahul Rajopadhye

(857)-376-9898 | rajopadhye.d@northeastern.edu | [linkedin.com/in/darshanrr](https://www.linkedin.com/in/darshanrr) | github.com/therrshan |
Boston, MA | Available Full-time: June 2025.

Education

Northeastern University

Boston, MA

Khoury College of Computer Sciences

Expected Graduation : May 2025

Master of Science in Artificial Intelligence

3.94 GPA

Relevant Coursework : Machine Learning, Deep Learning, Bias and Ethics in AI, Computer Vision.

Teaching Assistant : Machine Learning II (Prof. Chieh Wu)

Technical Skills

Programming Languages

Python, Java, C, C++, JavaScript, Bash, MATLAB, R.

Libraries & Frameworks

PyTorch, TensorFlow, HuggingFace, Scikit-Learn, LangChain, OpenCV, NLTK, Spacy.

Data Processing & Storage

Apache Spark, Nifi, Hadoop, BigQuery, Elasticsearch, FAISS, Pinecone, SQL, GraphQL

Software Development Tools

Flask, Django, REST APIs, Streamlit, Jira, Tableau, Power BI

Cloud & DevOps

AWS, GCP, Docker, Git, GitLab, MLflow, TensorBoard, Airflow, DVC.

Experience

Data Engineer

Aug 2021 – Sept 2022

Larsen and Toubro Infotech

Pune, IN

- Designed and optimized scalable ETL pipelines to process over 70GB of financial data daily, integrating structured and unstructured data to support advanced analytics and machine learning workflows.
- Developed automated data preprocessing and transformation pipelines using Python and Bash scripts, reducing manual intervention and enabling efficient feature extraction for downstream decision making.
- Designed high-performance storage solutions using AVRO and Parquet, enhancing data retrieval speeds.
- Implemented partitioning and indexing strategies in HiveDB, reducing query latency in batch processing workflows.

Projects

FixMatch Implementation | Python, PyTorch, Semi-Supervised Learning, CIFAR-10, [Link](#)

Oct 2024 - Dec 2024

- Reproduced FixMatch results on CIFAR-10 using Wide ResNet-28-2, achieving **90%** of the performance reported in the original paper.
- Implemented **weak** and **strong augmentations** with **pseudo-labeling**, utilizing **95%** unlabeled data for training.
- Optimized **hyperparameters** to closely align with the original paper, achieving performance within **5%**.
- Conducted extensive experiments, demonstrating the effectiveness of **semi-supervised learning** with **10%** higher accuracy than a fully supervised model.
- Evaluated various augmentation strategies, improving robustness by **7%** compared to standard methods.

Implicit Bias Annotation | Python, LLAMA-2, QLoRA, PEFT

Jan 2024 - May 2024

- Led cutting-edge research focused on developing NLP-based solutions for implicit bias detection, exploring novel techniques for automated bias annotation to assist in large-scale, nuanced evaluations of biases in textual data.
- Fine-tuned **LLAMA-2** model using **QLoRA methodology**, achieving state-of-the-art performance on the SBIC dataset.
- Implemented **PEFT** techniques to optimize the model's performance, significantly reducing computational costs while retaining high-quality results, making it feasible to scale.
- Developed a framework for annotating and classifying implicit bias using fine-tuned models, with potential applications in **content moderation**, fairness audits, and **bias monitoring**.

Pictophrases | Python, RNN, BiLSTM, Transformer, Image Captioning, Keras, [Link](#)

Sep 2023 – Dec 2023

- Conducted a study on advanced **image captioning** techniques using **RNNs**, **BiLSTMs**, and **Transformers**, achieving a **15%** improvement in BLEU score over baseline RNN models.
- Built the model architecture from scratch, incorporating **InceptionV3** for feature extraction.
- Evaluated model performance on the diverse **Flickr-8k dataset** using the **BLEU metric**, achieving an average score of **0.35**.
- Enhanced model accuracy through iterative training, hyperparameter tuning, and **dropout regularization**, improving **validation accuracy by 12%**.
- Implemented **beam search decoding** for generating more diverse captions, improving caption quality by **10%** according to human evaluation.